# A PROPHET SECRETARY INSTANCE WITH RATIO BELOW 0.7230

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Abstract. In the prophet secretary problem, independent nonnegative random variables with known distributions arrive in uniformly random order, and a gambler who sees each value as it arrives may stop once and keep it. The best constant $\alpha$ such that the gambler can always obtain $\alpha$ times the expected maximum lies between 0.688 and 0.7235, the upper bound being due to Giambartolomei, Mallmann-Trenn and Saona. We give an explicit instance with 461 variables on which the optimal gambler obtains less than 0.722934 times the expected maximum, so that $\alpha < 0.72294$. The instance consists of 400 rare prizes, two groups of 30 medium items with two different values, and one deterministic item. Its optimal value is computed exactly by a dynamic programme over counts of remaining items, evaluated in interval arithmetic. In the limit of many prizes the same family is described by an ordinary differential equation, whose numerical solution suggests that two medium values give about 0.72274 and three about 0.72268.

## 1. Introduction

Let $X_1, \dots, X_n$ be independent nonnegative random variables with known distributions. In the prophet secretary problem the variables are presented in a uniformly random order $\sigma$; at step $t$ the gambler learns the index $\sigma(t)$ and the value $X_{\sigma(t)}$, and either stops and receives $X_{\sigma(t)}$ or rejects it for good. Let $V(X_1, \dots, X_n)$ be the largest expected reward of a stopping rule adapted to this information, and let

$$\alpha = \inf \frac{V(X_1, \dots, X_n)}{\mathbb{E} \max_i X_i},$$

the infimum over all $n$ and all instances with $0 < \mathbb{E} \max_i X_i < \infty$.

When the order is fixed and adversarial the best constant is $1/2$, by the prophet inequality of Krengel and Sucheston [9] (see also Samuel-Cahn [10]). When the variables are identically distributed the order does not matter, and the best constant, studied by Hill and Kertz [8], is about 0.7451, as shown by Correa, Foncea, Hoeksma, Oosterwijk and Vredeveld [3]; in particular $\alpha \le 0.7451$.

Esfandiari, Hajiaghayi, Liaghat and Monemizadeh [6] showed $\alpha \ge 1 - 1/e$, Azar, Chiplunkar and Kaplan [1] were the first to improve on this constant, and the best lower bound is now 0.688, due to Chen, Huang, Li and Tang [4]. On the hardness side, Correa, Saona and Ziliotto [5] showed $\alpha \le \sqrt{3} - 1 \approx 0.732$, Bubna and Chiplunkar [2] obtained 0.7254 by a computer-assisted argument (their bound holds even for a gambler who knows the order in advance), and Giambartolomei, Mallmann-Trenn and Saona [7] proved $\alpha < 0.7235$ with an instance whose ratio they computed in closed form in a limit. We improve the upper bound.

**Theorem 1.1.** *There is an instance with* 461 *variables, each taking one positive value or* 0*, whose optimal prophet secretary value satisfies*

$$0.7229334600126004776 \le \frac{V}{\mathbb{E} \max_i X_i} \le 0.7229334600126004777.$$

*In particular* $\alpha < 0.72294$.

The instance, called $I(400, 30)$ below, has the shape of the instance of [7]. There are many independent prizes, each so rare that with high probability at most one of them appears, so that their total mass behaves like a stream of reward that arrives at a constant rate while the gambler

waits. There is one item that is certainly present and is worth less than the prize mass, and a cloud of medium items, each present with small probability, worth slightly more than the deterministic item. The gambler who meets the deterministic item early must decide whether to give up the remaining prize stream; the medium items make both waiting and stopping costly. In [7] the medium items all have one value. Taking two values, as anticipated in the conclusion of [7], lowers the ratio.

The value of the instance is computed by the standard recursion for the optimal stopping rule under random order (Section 2). Because items in a group are identically distributed, the recursion depends only on how many items of each group are left, which reduces the $2^{461}$ subsets to 770,722 states (Section 3). The recursion uses only arithmetic operations and maxima, so it can be evaluated in interval arithmetic with outward rounding, and the result is an enclosure of the exact ratio. Section 4 describes the limit of the family as the number of prizes grows; the values reported there are numerical and are not part of the theorem.

## 2. The recursion

For a set $S$ of indices let $V(S)$ be the optimal expected reward of the game played on the variables $(X_i)_{i\in S}$ alone, in uniformly random order, with $V(\emptyset) = 0$.

**Lemma 2.1.** *For nonempty $S$,*

$$V(S) = \frac{1}{|S|}\sum_{i\in S} \mathbb{E}\max\big(X_i, V(S\setminus\{i\})\big).$$

*Proof.* The first index presented is uniform on $S$. Given that it is $i$, the remaining order is uniform on the orderings of $S\setminus\{i\}$, and the remaining variables are independent of $X_i$ and of the order. Hence after observing $X_i$ the gambler who continues faces exactly the game on $S\setminus\{i\}$, whose optimal value is $V(S\setminus\{i\})$ whatever has been observed, and the optimal decision is to stop if and only if $X_i > V(S\setminus\{i\})$. Averaging over $i$ gives the formula; a backward induction over $|S|$ shows that the rule defined in this way is optimal among all stopping rules adapted to the observed indices and values. □

**Lemma 2.2.** *Suppose the index set is partitioned into groups $G_1,\dots,G_r$, the variables in each group being identically distributed, say $X_i \overset{d}{=} Y_g$ for $i\in G_g$. Then $V(S)$ depends on $S$ only through the counts $c_g = |S\cap G_g|$, and writing $V(c)$ for it,*

$$V(c) = \frac{1}{|c|}\sum_{g:\,c_g>0} c_g\,\mathbb{E}\max\big(Y_g, V(c-e_g)\big), \qquad |c| = c_1+\cdots+c_r, \quad V(0)=0.$$

*Proof.* Induction on $|S|$ using Lemma 2.1: the terms with $i\in S\cap G_g$ are all equal, since $X_i$ has the law of $Y_g$ and $S\setminus\{i\}$ has counts $c-e_g$. □

If $Y_g$ equals $v_g > 0$ with probability $\pi_g$ and 0 otherwise, then $\mathbb{E}\max(Y_g, w) = \pi_g\max(v_g, w) + (1-\pi_g)w$ for $w\ge 0$. If independent variables $X_i$ equal $v_i$ with probability $\pi_i$ and 0 otherwise, and $v_1\ge v_2\ge\cdots$, then

$$\mathbb{E}\max_i X_i = \sum_i v_i\pi_i\prod_{j<i}(1-\pi_j). \tag{1}$$

## 3. The instance and the certificate

Fix the parameters

$$a = 0.823, \quad b_1 = 1.28233, \quad p_1 = 0.2065, \quad b_2 = 1.38487, \quad p_2 = 0.17698, \quad q = 10^{-9},$$

and integers $k, m \ge 1$. The instance $I(k,m)$ has four groups:

- $k$ prizes, each equal to 1 with probability $q$ and to 0 otherwise;
- $m$ items, each equal to $B_1 = b_1 kq$ with probability $p_1/m$ and to 0 otherwise;
- $m$ items, each equal to $B_2 = b_2 kq$ with probability $p_2/m$ and to 0 otherwise;
- one item equal to $A = akq$.

All values of the medium and deterministic items are measured in units of the total prize mass $kq$; for $k = 400$ these values are of order $4 \cdot 10^{-7}$, far below the prize value 1. Since $1 > B_2 > B_1 > A$, (1) gives

$$\mathbb{E}\max = 1 - (1-q)^k + (1-q)^k \Big[ B_2\big(1 - (1 - \tfrac{p_2}{m})^m\big) + (1 - \tfrac{p_2}{m})^m \Big( B_1\big(1 - (1 - \tfrac{p_1}{m})^m\big) + (1 - \tfrac{p_1}{m})^m A \Big) \Big].$$

By Lemma 2.2 the optimal value is $V(k, m, m, 1)$, computed over the states $(j, l_1, l_2, s)$ with $0 \le j \le k$, $0 \le l_1, l_2 \le m$, $s \in \{0, 1\}$, of which there are $2(k+1)(m+1)^2$; for $k = 400$, $m = 30$ this is 770,722.

*Proof of Theorem 1.1.* Take the instance $I(400, 30)$, which has $400 + 30 + 30 + 1 = 461$ variables. The recursion of Lemma 2.2 was evaluated over all 770,722 states in interval arithmetic (the `iv` context of mpmath with a 100-bit mantissa), with each decimal parameter replaced by an interval containing it. Every operation in the recursion is an arithmetic operation or a maximum of two numbers; interval arithmetic with outward rounding returns an interval containing the exact result of an arithmetic operation applied to any points of its argument intervals, and $[x_1, y_1] \vee [x_2, y_2] = [\max(x_1, x_2), \max(y_1, y_2)]$ contains $\max(x, y)$ for every $x \in [x_1, y_1]$, $y \in [x_2, y_2]$. By induction over the states, each computed interval contains the exact value of its state. The same holds for the closed form of $\mathbb{E}\max$. The run returned

$$\frac{V}{\mathbb{E}\max} \in [\,0.722933460012600477658647124,\ 0.722933460012600477658647127\,],$$

which contains the stated interval. □

*Remark* 3.1. The computation takes about two and a half minutes on one core. Three further checks were made, none of which the proof uses. The recursion over counts agrees with the recursion of Lemma 2.1 over all $2^n$ subsets, run separately with every item distinct, for $I(6, 2)$ ($n = 11$, equal to the last bit in double precision) and $I(8, 3)$ ($n = 15$, difference $1.1 \cdot 10^{-16}$). Double precision gives 0.7229334600126 for $I(400, 30)$, in agreement with the certificate. Replacing $q = 10^{-9}$ by $q = 10^{-12}$ changes the ratio of $I(400, 30)$ by $7 \cdot 10^{-8}$, so the tiny prize probability enters only through the prize mass $kq$, as the construction intends.

At the same parameters the ratio decreases as the instance grows:

| $(k, m)$ | $(100, 10)$ | $(200, 20)$ | $(400, 30)$ | $(800, 40)$ |
|---|---|---|---|---|
| $V/\mathbb{E}\max$ | 0.7234029 | 0.7230763 | 0.7229335 | 0.7228621 |

(double precision). Every row is a valid instance; the theorem uses the third because it is the smallest one we certified. The successive differences roughly halve as $k$ doubles.

## 4. The many-prize limit

The following description of the limit is heuristic, and the numbers in this section are numerical solutions of an ordinary differential equation; they are not certified and are not claimed as bounds.

Let $k \to \infty$ with the prize mass $kq \to 0$, measure rewards in units of $kq$, and let the number of medium items of each value grow. Rescale time so that the game runs over $[0, 1]$. Then the prizes become a stream that pays reward at rate 1 for as long as the gambler waits (a single prize is always accepted, and to first order at most one appears), the medium items of value $b_j$ arrive as a Poisson process of rate $p_j$, and the deterministic item arrives at a uniform time. Writing $u$ for

the time remaining, the value $W(u)$ of the game after the deterministic item has passed and the value $\Phi(u)$ before it solve

$$W' = 1 + \sum_j p_j (b_j - W)^+, \quad W(0) = 0, \qquad \Phi' = 1 + \sum_j p_j (b_j - \Phi)^+ + \frac{\max(a, W) - \Phi}{u}, \quad \Phi(0) = a,$$

and the limiting ratio is $\Phi(1)$ divided by $1 + \mathbb{E}\max(a, Z)$, where $Z$ is the largest medium value that appears. With no medium items, $W(u) = u$ and $\Phi(1) = 1 + a^2/2$ for $a \le 1$, and the ratio $(1 + a^2/2)/(1 + a)$ is minimised at $a = \sqrt{3} - 1$ with value $\sqrt{3} - 1$, the instance of [5]. With one medium value this is the family of [7].

Solving these equations numerically (an adaptive Runge–Kutta method of order 8, relative tolerance $10^{-12}$, at parameters found by Nelder–Mead minimisation of the limit ratio) gives the following values.

| medium values | $a$ | $(b_j, p_j)$ | limit ratio |
|---|---|---|---|
| none | $\sqrt{3} - 1$ | | 0.73205081 |
| one, parameters of [7] | 0.789 | $(1.24, 0.421)$ | 0.72348603 |
| one, minimised | 0.8204 | $(1.3304, 0.3717)$ | 0.72302401 |
| two, parameters of $I(k, m)$ | 0.823 | $(1.28233, 0.2065)$, $(1.38487, 0.17698)$ | 0.72275062 |
| two, minimised | 0.82160 | $(1.27865, 0.20840)$, $(1.38431, 0.18833)$ | 0.72274261 |
| three, minimised | 0.82178 | $(1.26006, 0.14352)$, $(1.33157, 0.13374)$, $(1.40321, 0.12514)$ | 0.72268397 |

The second row agrees with the bound 0.7235 of [7]. The values of $I(k, m)$ in Section 3 approach the fourth row from above, the remaining gap at $k = 800$ being due to the finite number of medium items. With three values the optimal medium values are nearly evenly spaced, with nearly equal rates, which suggests that a continuous band of medium values is the extremal configuration within this family. The gains from adding a value decrease quickly ($2.8 \cdot 10^{-4}$ from one to two, $5.9 \cdot 10^{-5}$ from two to three).

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## Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

## Declaration of generative AI and AI-assisted technologies

Anthropic Claude AI systems were used to develop the mathematical derivations, write and run the numerical checks, perform internal consistency and literature reviews, and write and prepare this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for submission

or publication rests with the author. The AI-assisted internal reviews, compilation checks and rendered-page inspections are not external peer review or formal verification.

## Data and code availability

The certificate of Theorem 1.1, the checks of Section 3 and the numerical solutions of Section 4 are produced by the scripts in the `code` directory of the repository

`github.com/yspennstate/prophet-secretary-hardness`.

The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [11].

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